Dataset 2: TV TRP

Table name: tv_trp

Operation 1: Filtering with WHERE

Question: Which programme records have a TRP of 6.0 or higher?

```
SELECT Month, Programme, Time_Slot, TRP, Viewers_Lakh
FROM tv_trp
WHERE TRP >= 6.0
ORDER BY TRP DESC;
```

What it demonstrates: Finding high-performing programme records using a condition.

Operation 2: Sorting with ORDER BY and LIMIT

Question: Which five records have the highest viewer counts?

```
SELECT Month, Programme, Region, Viewers_Lakh
FROM tv_trp
ORDER BY Viewers_Lakh DESC
LIMIT 5;
```

What it demonstrates: Ranking programme records by audience size.

Operation 3: Aggregation with GROUP BY

Question: Which time slots have the highest average TRP?

```
SELECT Time_Slot,
    COUNT(*) AS ProgrammeRecords,
    ROUND(AVG(TRP), 2) AS AverageTRP,
    ROUND(AVG(Viewers_Lakh), 2) AS AverageViewers
FROM tv_trp
GROUP BY Time_Slot
ORDER BY AverageTRP DESC;
```

What it demonstrates: Comparing performance across morning, afternoon, and evening time slots.

Operation 4: Filtering groups with HAVING

Question: Which programmes have an average TRP greater than 5.5?

```
SELECT Programme,
       ROUND(AVG(TRP), 2) AS AverageTRP,
       COUNT(*) AS Records
FROM tv_trp
GROUP BY Programme
HAVING AVG(TRP) > 5.5
ORDER BY AverageTRP DESC;
```

What it demonstrates: Filtering programmes based on an aggregate measure.

Operation 5: Subquery

Question: Which programme records have a TRP above the overall dataset average?

```
SELECT Month, Programme, TRP
FROM tv_trp
WHERE TRP > (
   SELECT AVG(TRP)
   FROM tv_trp
)
ORDER BY TRP DESC;
```

What it demonstrates: Comparing each programme record with the overall average TRP calculated by a subquery.

Operation 6: Self-join

Question: How did each programme's TRP change between January and June 2026?

```
SELECT jan.Programme,
       jan.TRP AS JanuaryTRP,
       jun.TRP AS JuneTRP,
       ROUND(jun.TRP - jan.TRP, 2) AS TRPChange
FROM tv_trp AS jan
JOIN tv_trp AS jun
   ON jan.Programme = jun.Programme
WHERE jan.Month = '2026-01'
  AND jun.Month = '2026-06'
ORDER BY TRPChange DESC;
```

What it demonstrates: Joining a table to itself to compare two periods for the same programme. Only programmes with records in both months will appear.

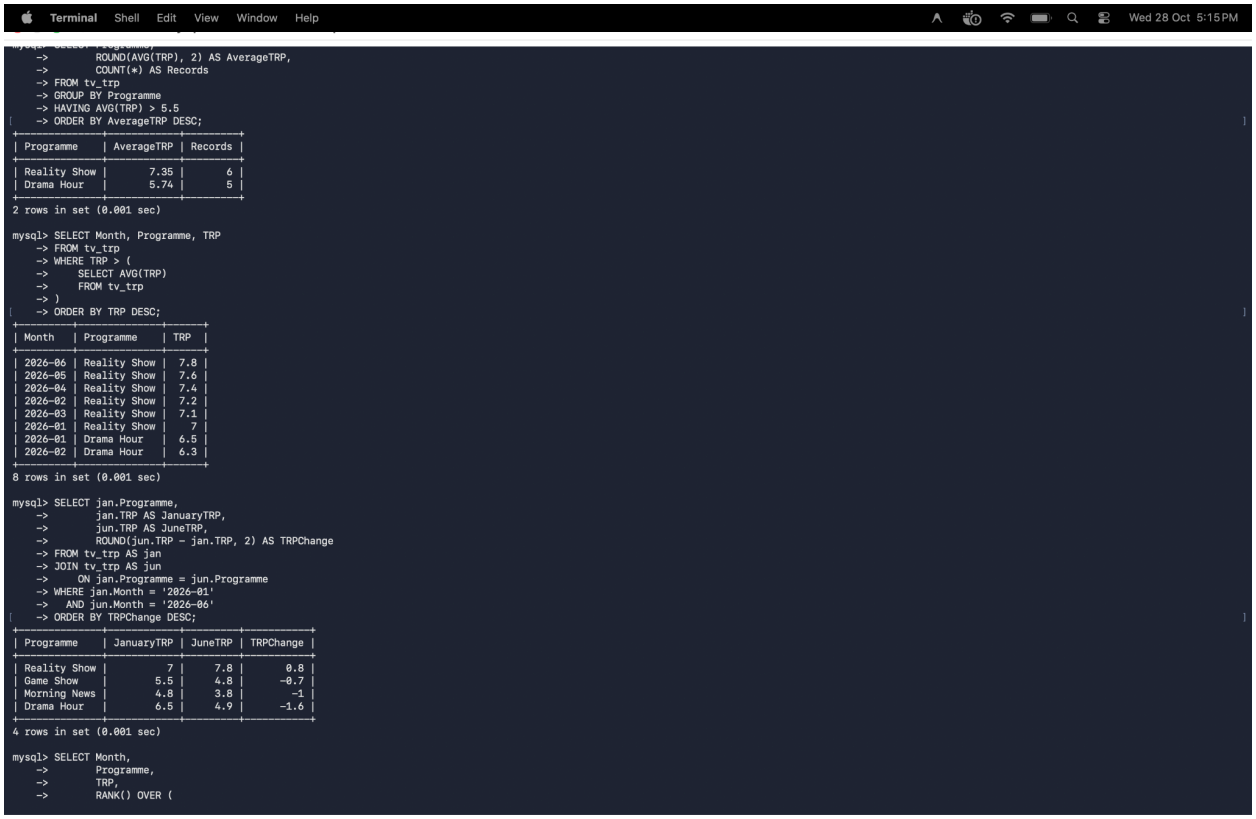
Operation 7: Window function with RANK()

Question: What is each programme's TRP ranking within its month?

```
SELECT Month,
       Programme,
       TRP,
       RANK() OVER (
          PARTITION BY Month
          ORDER BY TRP DESC
       ) AS TRPRank
FROM tv_trp
ORDER BY Month, TRPRank;
```

What it demonstrates: Ranking programmes separately within each month without collapsing the individual rows.

Results



```
| Drama Hour  |       6.5 |     4.9 |      -1.6 |
+-------------+------------+---------+-----------+
4 rows in set (0.001 sec)

mysql> SELECT Month,
    ->        Programme,
    ->        TRP,
    ->        RANK() OVER (
    ->            PARTITION BY Month
    ->            ORDER BY TRP DESC
    ->        ) AS TRPRank
    -> FROM tv_trp
    -> ORDER BY Month, TRPRank;
+---------+--------------+------+---------+
| Month   | Programme    | TRP  | TRPRank |
+---------+--------------+------+---------+
| 2026-01 | Reality Show |    7 |       1 |
| 2026-01 | Drama Hour   |  6.5 |       2 |
| 2026-01 | Game Show    |  5.5 |       3 |
| 2026-01 | Morning News |  4.8 |       4 |
| 2026-02 | Reality Show |  7.2 |       1 |
| 2026-02 | Drama Hour   |  6.3 |       2 |
| 2026-02 | Game Show    |  5.6 |       3 |
| 2026-02 | Morning News |  4.6 |       4 |
| 2026-03 | Reality Show |  7.1 |       1 |
| 2026-03 | Game Show    |  5.4 |       2 |
| 2026-04 | Reality Show |  7.4 |       1 |
| 2026-04 | Drama Hour   |  5.7 |       2 |
| 2026-04 | Game Show    |  5.2 |       3 |
| 2026-04 | Morning News |  4.2 |       4 |
| 2026-05 | Reality Show |  7.6 |       1 |
| 2026-05 | Drama Hour   |  5.3 |       2 |
| 2026-05 | Game Show    |    5 |       3 |
| 2026-05 | Morning News |    4 |       4 |
| 2026-06 | Reality Show |  7.8 |       1 |
| 2026-06 | Drama Hour   |  4.9 |       2 |
| 2026-06 | Game Show    |  4.8 |       3 |
| 2026-06 | Morning News |  3.8 |       4 |
+---------+--------------+------+---------+
22 rows in set (0.002 sec)

mysql> SELECT Programme,
    ->        ROUND(AVG(TRP), 2) AS ChannelTRP,
    ->        ROUND(AVG(Competitor_TRP), 2) AS CompetitorTRP,
    ->        ROUND(AVG(TRP - Competitor_TRP), 2) AS AverageGap
    -> FROM tv_trp
    -> GROUP BY Programme
    -> ORDER BY AverageGap DESC;
+--------------+------------+---------------+------------+
| Programme    | ChannelTRP | CompetitorTRP | AverageGap |
+--------------+------------+---------------+------------+
| Reality Show |       7.35 |          7.92 |      -0.57 |
| Game Show    |       5.25 |          5.82 |      -0.57 |
| Morning News |       4.28 |          4.88 |       -0.6 |
| Drama Hour   |       5.74 |          6.34 |       -0.6 |
+--------------+------------+---------------+------------+
4 rows in set (0.001 sec)

mysql>
```

```
mysql>  SELECT * FROM tv_trp LIMIT 5;
+---------+--------------+-----------+------+-------------+----------------+-----------+--------+-----------------+
| Month   | Programme    | Time_Slot | TRP  | Viewers_Lakh | Competitor_TRP | Age_Group | Region | Ad_Revenue_Lakh |
+---------+--------------+-----------+------+-------------+----------------+-----------+--------+-----------------+
| 2026-01 | Morning News | Morning   |  4.8 |          7.2 |            5.1 | 18–24     | North  |              12 |
| 2026-02 | Morning News | Morning   |  4.6 |          6.9 |            4.8 | 25–34     | South  |            11.5 |
| 2026-04 | Morning News | Morning   |  4.2 |          6.3 |            4.9 | 45+       | West   |            10.5 |
| 2026-05 | Morning News | Morning   |    4 |            6 |            4.8 | 18–24     | North  |              10 |
| 2026-06 | Morning News | Morning   |  3.8 |          5.7 |            4.8 | 25–34     | South  |             9.5 |
+---------+--------------+-----------+------+-------------+----------------+-----------+--------+-----------------+
5 rows in set (0.001 sec)

mysql> SELECT Month, Programme, Time_Slot, TRP, Viewers_Lakh
    -> FROM tv_trp
    -> WHERE TRP >= 6.0
    -> ORDER BY TRP DESC;
+---------+--------------+-----------+------+--------------+
| Month   | Programme    | Time_Slot | TRP  | Viewers_Lakh |
+---------+--------------+-----------+------+--------------+
| 2026-06 | Reality Show | Prime Time |  7.8 |         11.7 |
| 2026-05 | Reality Show | Prime Time |  7.6 |         11.4 |
| 2026-04 | Reality Show | Prime Time |  7.4 |         11.1 |
| 2026-02 | Reality Show | Prime Time |  7.2 |         10.8 |
| 2026-03 | Reality Show | Prime Time |  7.1 |        10.65 |
| 2026-01 | Reality Show | Prime Time |    7 |         10.5 |
| 2026-01 | Drama Hour   | Prime Time |  6.5 |         9.75 |
| 2026-02 | Drama Hour   | Prime Time |  6.3 |         9.45 |
+---------+--------------+-----------+------+--------------+
8 rows in set (0.003 sec)

mysql> SELECT Month, Programme, Region, Viewers_Lakh
    -> FROM tv_trp
    -> ORDER BY Viewers_Lakh DESC
    -> LIMIT 5;
+---------+--------------+--------+--------------+
| Month   | Programme    | Region | Viewers_Lakh |
+---------+--------------+--------+--------------+
| 2026-06 | Reality Show | South  |         11.7 |
| 2026-05 | Reality Show | North  |         11.4 |
| 2026-04 | Reality Show | West   |         11.1 |
| 2026-02 | Reality Show | South  |         10.8 |
| 2026-03 | Reality Show | East   |        10.65 |
+---------+--------------+--------+--------------+
5 rows in set (0.000 sec)

mysql> SELECT Time_Slot,
    ->        COUNT(*) AS ProgrammeRecords,
    ->        ROUND(AVG(TRP), 2) AS AverageTRP,
    ->        ROUND(AVG(Viewers_Lakh), 2) AS AverageViewers
    -> FROM tv_trp
    -> GROUP BY Time_Slot
    -> ORDER BY AverageTRP DESC;
+------------+------------------+------------+-----------------+
| Time_Slot  | ProgrammeRecords | AverageTRP | AverageViewers |
+------------+------------------+------------+-----------------+
| Prime Time |               11 |       6.62 |            9.93 |
| Evening    |                6 |       5.25 |            7.88 |
| Morning    |                5 |       4.28 |            6.42 |
+------------+------------------+------------+-----------------+
3 rows in set (0.002 sec)
```